

NAME: KARUNYA M A

ROLL NO:22ISR016

Kubernetes

Kubernetes is an open-source container orchestration system for automating the deployment, scaling, and management of containerized applications. It was originally developed by Google and is now maintained by the Cloud Native Computing Foundation (CNCF).

1.Installing Kubernetes

```
root@LAPTOP-N670L8SV:/home/karunymathivanan# curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 138 100 138    0     0  182      0 --:--:-- --:--:-- --:--:-- 182
100 54.6M 100 54.6M    0     0 2876k      0 0:00:19 0:00:19 --:--:-- 4454k
root@LAPTOP-N670L8SV:/home/karunymathivanan# chmod +x kubectl
root@LAPTOP-N670L8SV:/home/karunymathivanan# sudo mv kubectl /usr/local/bin/
root@LAPTOP-N670L8SV:/home/karunymathivanan# kubectl version --client
Client Version: v1.32.1
Kustomize Version: v5.5.0
```

2.Starting minikube

```
root@LAPTOP-N670L8SV:/home/karunymathivanan# minikube start
🐳 minikube v1.35.0 on Ubuntu 24.04 (amd64)
🔧 Automatically selected the docker driver. Other choices: none, ssh
🚨 The "docker" driver should not be used with root privileges. If you wish to continue as root, use --force.
💡 If you are running minikube within a VM, consider using --driver=none:
🔗 https://minikube.sigs.k8s.io/docs/reference/drivers/none/
```

3.Create a deployment : `kubectl create deployment dev --image=karunya08/pushimages --port=80`

4.Expose a deployment: `kubectl expose deployment dev --type=NodePort --port=80`

5.get deployments : `kubectl get deployments`

```
exit
karunymathivanan@LAPTOP-N670L8SV:~$ kubectl create deployment dev --image=karunya08/pushimages --port=80
deployment.apps/dev created
karunymathivanan@LAPTOP-N670L8SV:~$ kubectl expose deployment dev --type=NodePort --port=80
service/dev exposed
karunymathivanan@LAPTOP-N670L8SV:~$ kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
dev     0/1     1            0           33s
```

6.get pods : `kubectl get pods`

7. The minikube service <service-name> command is used to access a Kubernetes service running in a Minikube cluster.

Command: **minikube service dev**

```
karunymathivanan@LAPTOP-N670L8SV:~$ kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
dev 1/1 1 1 56s
karunymathivanan@LAPTOP-N670L8SV:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
dev-7bcb684f4c-xzhg5 1/1 Running 0 63s
karunymathivanan@LAPTOP-N670L8SV:~$ minikube service dev
```

NAMESPACE	NAME	TARGET PORT	URL
default	dev	80	http://192.168.49.2:31255

```

🚀 Starting tunnel for service dev.

```

NAMESPACE	NAME	TARGET PORT	URL
default	dev		http://127.0.0.1:46075

```

🌐 Opening service default/dev in default browser...
👉 http://127.0.0.1:46075
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.

```

8.Output

Welcome to Jenkins and
docker

This is a basic HTML page with some CSS styling